

Describe translations

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Point P moves from (2, -5) to (-3, 1). Describe the translation. Copy or complete the first step.

2. Translate (4, -2) by 3 left and 5 up.

3. Reflect (-6, 2) in the y-axis.

4. Miro says: Miro changes x and y in the same way without checking the direction or mirror line. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Point P moves from (2, -5) to (-3, 1). Describe the translation. Copy or complete the first step.
Answer 5 left and 6 up.
2. Translate (4, -2) by 3 left and 5 up.
Answer (1, 3).
3. Reflect (-6, 2) in the y-axis.
Answer (6, 2).
4. Miro says: Miro changes x and y in the same way without checking the direction or mirror line. Is this correct? Explain.
Answer Track horizontal and vertical changes separately and apply the exact transformation to every point.

Describe translations

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Point P moves from (2, -5) to (-3, 1). Describe the translation.

6. Check this result using an inverse, estimate or known fact: Reflect (-6, 2) in the y-axis.

2. Translate (4, -2) by 3 left and 5 up.

7. Prove or explain your reasoning: A point becomes (7, -4) after moving 9 right and 3 down. Find its start.

3. Reflect (-6, 2) in the y-axis.

8. Identify and correct this misconception: Miro changes x and y in the same way without checking the direction or mirror line.

4. Solve this independently and show every stage: Point P moves from (2, -5) to (-3, 1). Describe the translation.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Translate (4, -2) by 3 left and 5 up.

10. Write and solve a similar question to: Reflect (-6, 2) in the y-axis.

Teacher answers and checking prompts

1. Point P moves from (2, -5) to (-3, 1). Describe the translation.
Answer 5 left and 6 up.
2. Translate (4, -2) by 3 left and 5 up.
Answer (1, 3).
3. Reflect (-6, 2) in the y-axis.
Answer (6, 2).
4. Solve this independently and show every stage: Point P moves from (2, -5) to (-3, 1). Describe the translation.
Answer 5 left and 6 up.
5. Use a second method or representation for: Translate (4, -2) by 3 left and 5 up.
Answer Expected result: (1, 3). Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Reflect (-6, 2) in the y-axis.
Answer Expected result: (6, 2). A valid check must be shown.
7. Prove or explain your reasoning: A point becomes (7, -4) after moving 9 right and 3 down. Find its start.
Answer (-2, -1). The explanation must show why the method works.
8. Identify and correct this misconception: Miro changes x and y in the same way without checking the direction or mirror line.
Answer Track horizontal and vertical changes separately and apply the exact transformation to every point.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Reflect (-6, 2) in the y-axis.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Describe translations

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: A point becomes (7, -4) after moving 9 right and 3 down. Find its start.

6. Create a new example that tests the same idea as: Reflect (-6, 2) in the y-axis.

2. Prove the answer to this question, rather than only calculating it: Point P moves from (2, -5) to (-3, 1). Describe the translation.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Translate (4, -2) by 3 left and 5 up.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: (6, 2).

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro changes x and y in the same way without checking the direction or mirror line.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: A point becomes (7, -4) after moving 9 right and 3 down. Find its start.
Answer (-2, -1). The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Point P moves from (2, -5) to (-3, 1). Describe the translation.
Answer Expected result: 5 left and 6 up. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Translate (4, -2) by 3 left and 5 up.
Answer Expected result: (1, 3). Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: (6, 2).
Answer One valid reconstruction is: Reflect (-6, 2) in the y-axis. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro changes x and y in the same way without checking the direction or mirror line.
Answer Track horizontal and vertical changes separately and apply the exact transformation to every point.
6. Create a new example that tests the same idea as: Reflect (-6, 2) in the y-axis.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.